

PRACTICAL EXAM TRAINER WORKSHEET WATER & PREVENTIVE MAINTENANCE FOUNDATION COFFEE TECHNICIANS PROGRAM

實作測驗 培訓師工作表

水與預防性維護基礎課程

咖啡技師學程

This optional trainer worksheet is designed to help a trainer efficiently proctor, observe, and score the practical exam for up to six (6) learners based on the instructions within the Trainer Guide. Trainers may also use other methods to record the individual points for the different tasks included in the exam. Following the completion of the practical exams, trainers will need to transfer point totals to the appropriate spaces on the Practical Exam Learner Worksheet for each student.

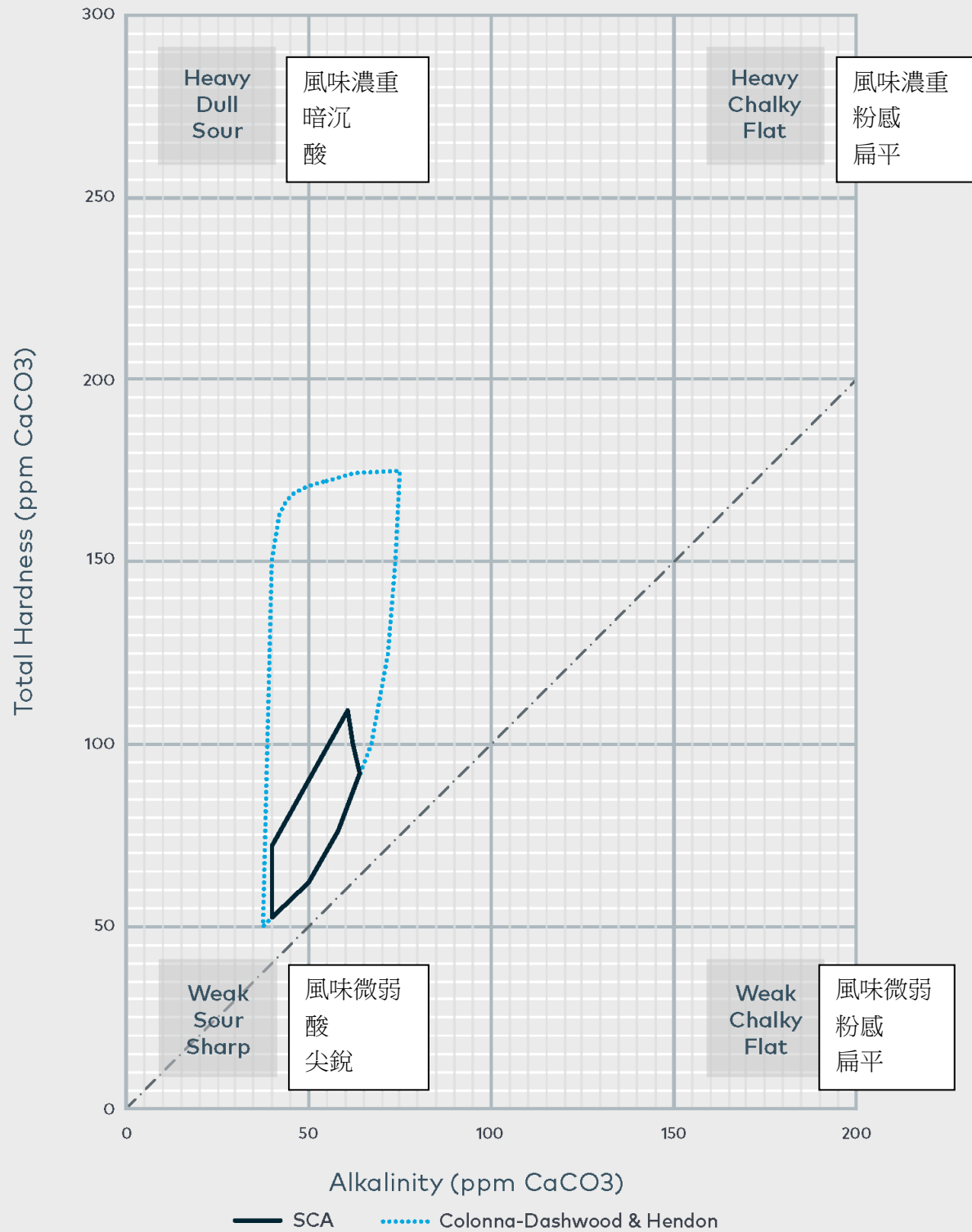
本工作表（非必要）的設計旨在協助培訓師透過有效率的方式，為最多六（6）名學員監考、觀察和評量實作測驗。培訓師也可透過其他方式，記錄學員在測驗中各項任務的分數。完成施測後，培訓師應將成績加總，並妥善註記在每個學員的『實作測驗學員工作表』上。

施測日期 Date of Course: _____

LEARNERS: 學習者:

1.
2.
3.
4.
5.
6.

Use this chart as a answer key for Section 1, Skill B. 使用此圖表作為第一部分、技能 B 之解答參考。



Section 1 – Measuring Water Composition

第一部分 – 水中成分量測

Skill A - Measure the Total Hardness and Alkalinity of a water sample

技能 A – 水樣總硬度與鹼度之量測

Task	Pts						
Learner properly collects samples and fills testing vials 學習者正確地蒐集樣本並填充試瓶。	1						
Learner uses appropriate technique for adding reagent drops and agitating sample (1pt per vial) 學習者使用適當的技巧，於水樣中添加試劑並進行攪拌。(每個試瓶 1 分)	2						
Learner correctly recognizes color change and drops added for Total Hardness 學員正確地辨認測量總硬度時的顏色變化及添加的滴定量。	1						
Learner correctly recognizes color change and drops added for Alkalinity 學員正確地辨認測量鹼度時的顏色變化及添加的滴定量。	1						
Learner correctly records measurements on Learner Worksheet 學員正確地將測量結果記錄在學員工作表。	1						
Total	6						

Skill B - Record Total Hardness and Alkalinity measurements on the SCA Water Chart

技能 B-在 SCA 水質圖記錄總硬度及鹼度測量值

Task	Pts						
Learner correctly performs calculation to convert measurements into ppm CaCO ₃ (1pt per value) 學員正確地執行運算將測量結果轉換成 ppm CaCO ₃ 值。(每個數值一分)	2						
Learner accurately plots the Total Hardness result obtained on the provided chart 學員準確地在提供的圖表中標示出所獲得的總硬度結果。	2						
Learner accurately plots the Alkalinity result obtained on the provided chart	2						

學員準確地在提供的圖表中標示出所獲得的鹼度結果。							
Total	6						

Skill C - Use an Electrical Conductivity meter to measure the estimated TDS of a water sample

技能 C-使用導電度計測量水質樣本的 TDS 概略值。

Task 任務	Pts 分數						
Learner properly collects water sample 學員正確地收集水質樣本。	1						
Learner uses appropriate technique for using electrical conductivity meter 學員使用適當的技巧操作導電度計。	1						
Learner correctly records estimated TDS measurement on Learner Worksheet 學員正確地將 TDS 測量結果記錄在學員工作表。	1						
Total 總分	3						

Trainer Comments: 講師評語

Section 2 - Water Treatment System Maintenance

Replace a cartridge in a drop-in style water filter system

更換置入式過濾系統的濾芯

Task	Pts						
Learner turns off water supply to water filter 學員關閉濾水器的供水。	2						
Learner removes filter housing and old water filter cartridge 學員拆下濾水器外殼並移除老舊濾芯。	2						
Learner properly cleans and sanitizes filter housing 學員正確地清潔並消毒濾水器外殼。	2						
Learner installs new water filter cartridge and reassembles housing properly 學員安裝新的濾芯並正確地重新組裝外殼。	2						
Learner turns on water supply to filter and flushes system appropriately 學員打開濾水器的供水並適當地沖洗過濾系統。	2						
Filter system does not leak 過濾系統不會漏水。	2						
Learner records last change date and replacement on Learner Worksheet 學員在學員工作表上記錄最後更換日期及更換項目。	1						
Total 總分	13						

Trainer Comments:

Section 3 - Coffee Brewer and Espresso Grinder Preventive Maintenance

第三章-咖啡沖煮機及義式磨豆機預防性維護

Skill A - Clean a coffee batch brewer level probe

技能 A- 清潔滴濾式咖啡機的水位探針

Task	Pts						
Learner confirms that coffee brewer is disconnected from electrical power 學員確認滴濾式咖啡機未接上電源。	2						
Learner removes required access panel and places screws in a secure location 學員拆除所需的檢修孔蓋並將螺絲放置在妥善的位置。	2						
Learner removes level probe without damaging probe or mounting bushings 學員在不損害探針或安裝套管的情況下拆除水位探針。	2						
Learner uses appropriate means to fully clean the level probe 學員使用適當的方法完整地清潔水位探針。	2						
Learner reinstalls level probe and connects wires if required 學員重新安裝水位探針並根據需求接上電線。	2						
Learner reassembles access panel and tightens screws appropriately 學員重新安裝檢修孔蓋並適當地栓緊螺絲。	2						
Learner records activity completion on Learner Worksheet 學員將活動完成情形記錄在學員工作表上。	1						
Total	13						

Skill B - Replace the burrs in an espresso grinder

技能 3B-更換義式磨豆機的刀盤

Task	Pts						
Learner unplugs espresso grinder before beginning service 學員在開始維護前拔除義式磨豆機的插頭。	2						
Learner opens grinding chamber and removes worn grinding burrs 學員拆開研磨腔室並移除磨損的刀盤。	3						
Learner thoroughly cleans grinding chamber and burr mounting surfaces 學員徹底地清潔研磨腔室及刀盤的安裝面。	2						
Learner installs new grinder burrs, using proper screw tightening sequence 學員使用合適的螺絲拴緊方式安裝新的刀盤。	3						
Learner reassembles grinder and connects to electrical outlet 學員重新安裝磨豆機並接上插座。	3						
Learner tests for proper function and adjusts to approximately correct setting 學員測試功能是否正常並調整至大略正確的設定。	2						
Learner records activity completion on Learner Worksheet 學員將活動完成情形記錄在學員工作表上。	1						
Total	16						

Trainer Comments:

Section 4 - Espresso Machine Preventive Maintenance 義式咖啡機的預防性維護

Skill A - Replace an espresso machine group screen and gasket

技能 A- 更換義式咖啡機的沖煮頭分水網和墊圈

Task	Pts						
Learner properly removes group screen and screw 學員正確地拆除沖煮頭分水網及螺絲。	2						
Learner properly removes group gasket without damaging brewing group 學員正確地拆除墊圈且未損壞沖煮頭。	3						
Learner cleans group gasket channel and group 學員將墊圈溝槽及沖煮頭清潔乾淨。	2						
Learner orients group gasket correctly when installing 學員以正確的一面定位墊圈。	2						
Learner properly installs new group gasket 學員正確地安裝新墊圈。	2						
Learner installs group screen and screw correctly, using appropriate screw torque 學員正確地安裝沖煮頭分水網及螺絲，並使用適當的力道拴緊螺絲。	2						
Learner records activity completion on Learner Worksheet 學員將活動完成情形記錄在學員工作表上。	1						
Total	14						

Skill B - Rebuild an espresso machine steam wand joint

技能 B-重新組裝一只義式咖啡機的蒸氣棒接頭

Task	Pts						
Learner properly removes steam wand from steam valve 學員正確地從蒸氣閥上拆下蒸氣棒。	2						
Learner properly removes wear parts as directed without damage to equipment 學員根據引導正確地拆除磨損零件且未對設備造成損壞。	2						
Learner cleans steam wand joint area and valve as directed	2						

學員根據引導將蒸氣棒接頭區塊及蒸汽閥清潔乾淨。							
Learner properly installs replacement parts, applying lubricant where applicable 學員正確地安裝更換零件，並在適當的地方添加潤滑劑。	3						
Learner properly reassembles steam wand to steam valve 學員正確地將蒸氣棒重新安裝回蒸氣閥。	2						
Steam wand joint does not leak when tested 確認測試時蒸氣棒接頭不會洩漏。	2						
Learner records activity completion on Learner Worksheet 學員將活動完成情形記錄在學員工作表上。	1						
Total 總分	14						

Skill C - Complete an espresso machine inspection

技能 C-完成義式咖啡機的檢查

Task	Pts 分數						
Learner correctly records steam boiler pressure on Learner Worksheet 學員正確地在學員工作表上記錄蒸氣鍋爐壓力。	1						
Learner observes pump pressure using blank portafilter basket or thermofilter device 學員透過使用無孔濾杯或測溫測壓把手觀察泵浦壓力。	2						
Learner correctly records pump pressure on Learner Worksheet 學員正確地將蒸氣泵浦壓力記錄在學員工作表上。	1						
Learner tests anti-vacuum valve and records operating state on Learner Worksheet 學員測試真空排除閥並將運作狀況記錄在學員工作表上。	2						

Learner inspects safety valve and records operating state on Learner Worksheet 學員檢查安全閥並將運作狀況記錄在學員工作表上。	1						
Learner tests drain function and records operating state on Learner Worksheet 學員測試排水功能並將運作狀況記錄在學員工作表上。	2						
Learner inspects machine for water leaks and records operating state on Learner Worksheet 學員檢查機器是否漏水並將運作狀況記錄在學員工作表上。	2						
Learner inspects machine for steam leaks and records operating state on Learner Worksheet 學員檢查機器的蒸氣是否會洩漏並將狀況記錄在學員工作表上。	2						
Learner measures group water debit and records observations on Learner Worksheet 學員測量沖煮頭出水量，並將觀測結果記錄於學員工作表	2						
Total 總分	15						

Trainer Comments: 講師評語

Use this copy of the Preventive Maintenance Checklist as an answer key for the exam.

Practical Exam - Preventive Maintenance Checklist

實務測驗-預防性維護核對表

Water Filtration System 水過濾系統		Service Required? 需要維護?	Service Complete 維護完成
Service or Inspection Item 維護或檢查項目	Observation 觀察結果		
Measure Total Hardness 測量總硬度		Yes / No 是/否	
Measure Alkalinity 測量鹼度		Yes / No 是/否	
Measure TDS 測量 TDS		Yes / No 是/否	
Service water filter cartridge (record last changed date) 更換濾水器濾芯(記錄最後更換日期)			
Traditional Espresso Machine 傳統義式咖啡機		Service Required? 需要維護?	Service Complete 維護完成
Service or Inspection Item 維護或檢查項目	Observation 觀察結果		
Clean brewing group 清潔沖煮頭			
Replace group gaskets 更換沖煮頭墊圈			
Replace group screens 更換沖煮頭分水網			
Rebuild steam wand joint 重新組裝蒸氣棒接頭			
Inspect for steam valve and steam wand joint leaks 檢查蒸氣閥及蒸汽棒接頭是否洩漏		Yes / No 是/否	
Record steam boiler pressure 記錄蒸氣鍋爐壓力		Yes / No 是/否	
Record pump pressure 記錄泵浦壓力		Yes / No 是/否	
Inspect steam boiler safety valve 檢查蒸氣鍋爐安全閥		Yes / No 是/否	

Test anti-vacuum valve function 測試真空排除閥功能		Yes / No 是/否	
Inspect expansion valve 檢查膨脹閥		Yes / No 是/否	
Test auto fill system function / steam boiler water level 測試自動補水系統功能/蒸氣鍋爐水位		Yes / No 是/否	
Test machine drain function 測試機器的排水功能		Yes / No 是/否	
Inspect for water leaks overall 徹底檢查是否漏水		Yes / No 是/否	
Inspect for steam leaks overall 徹底檢查蒸氣是否洩漏		Yes / No 是/否	
Coffee Batch Brewer 滴濾式咖啡機		Service Required? 需要維護?	Service Complete 維護完成
Service or Inspection Item 維護或檢查項目	Observation 觀察結果		
Clean level probe (record any build-up) 清潔水位探針(記錄任何結垢)			
Espresso Grinder 義式磨豆機		Service Required? 需要維護?	Service Complete 維護完成
Service or Inspection Item 維護或檢查項目	Observation 觀察結果		
Clean grinding chamber and replace burrs 清潔研磨腔室並更換刀盤			
Adjust grind setting 調整研磨設定			